

Hemoglobin Thionville

AN α -CHAIN VARIANT WITH A SUBSTITUTION OF A GLUTAMATE FOR VALINE AT NA-1 AND HAVING AN ACETYLATED METHIONINE NH₂ TERMINUS*

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In hemoglobin (Hb) Thionville, the substitution of a glutamic acid for the α -chain NH₂-terminal valine inhibits the cleavage of the initiator methionine which is then acetylated. The elongation of the α -chain NH₂ terminus modifies the three-dimensional structure of hemoglobin at a region that is known to have an important role in the allosteric regulation of oxygen binding. Relative to Hb A, Hb Thionville has a lower affinity for oxygen, and the heterotropic allosteric effects of protons, chloride, and bezafibrate are reduced. In contrast, the response to 2,3-diphosphoglycerate is normal. Analysis of oxygen equilibrium data within the framework of the two-state allosteric model indicates that the structure of deoxy Hb Thionville is stabilized relative to that of deoxy Hb A. The x-ray crystal structure of deoxy Hb Thionville shows that the glutamate side chain extends away from the $\alpha 1$ - $\alpha 2$ interface, whereas the methionine side chain (which has two conformations) extends into the $\alpha 1$ - $\alpha 2$ interface, physically displacing chloride and bezafibrate. The increased stability of deoxy Hb Thionville is due to new intrasubunit and intersubunit contacts made by the methionine. These interactions replace the indirect contacts, made through bound chloride ions, that Val-1 α normally contributes to the $\alpha 1$ - $\alpha 2$ interface.

In humans, globin translation is initiated with an unblocked methionine residue. When the nascent chain reaches a length of about 20–30 amino acid residues, the normal processing involves the release of the initiator methionine through the action of a methionine aminopeptidase. The release of the initiator residue is inhibited when a charged amino acid or one with a bulky side chain follows the methionine (Boissel *et al.*, 1988). In the β -chain, as a consequence of such a mechanism, several hemoglobin variants in which the initiator methionine is retained have been described. In Hb¹ South

Florida (Boissel *et al.*, 1985) and Hb Doha (Kamel *et al.*, 1985) the NA1-valine is replaced, respectively, by methionine and by glutamic acid. In Hb Long Island-Marseille (Blouquit *et al.*, 1984; Barwick *et al.*, 1985), the amino acid substitution concerns the second residue of the β -chain, which is a proline instead of a histidine.

Additional processing can occur when the chain reaches a length of 40–50 residues. This modification consists of the cotranslational acetylation of NH₂-terminal residue. The efficiency of N^α-acetylation depends on the NH₂-terminal structure of the protein; a NH₂-terminal valine strongly inhibits acetylation (Boissel *et al.*, 1988).

We report here Hb Thionville, the first example of an alteration of the NH₂ terminus of the α -chain. In Hb Thionville the NH₂-terminal valine of the α -chain is replaced by a glutamic acid, and this results in the retention and acetylation of the initiator methionine. The effect of the Hb Thionville mutation on cotranslational processing is identical to that seen with Hb Doha, except that the replacement of the NH₂-terminal valine with a glutamate-acetylated methionine occurs on the β -chain in Hb Doha (Kamel *et al.*, 1985).

Mutation and elongation of the α -chain NH₂ termini in hemoglobin Thionville alters the structure of a region of the hemoglobin tetramer that is intimately involved in the regulation of oxygen binding properties. The NH₂ terminus of the α -chain participates indirectly, through bound solvent interactions, in a network of polar and charge contacts that includes the COOH terminus of the opposite α -chain (Fermi, 1975; O'Donnell *et al.*, 1979). These interactions, which are not present in liganded hemoglobin, are an essential element of the allosteric mechanism (Perutz, 1989). Disruption of these interactions, either by mutation (Poyart *et al.*, 1980), enzymatic modification (Antonini *et al.*, 1961; Kilmartin *et al.*, 1975, 1977), or chemical modification (Kilmartin *et al.*, 1973; Nigen *et al.*, 1980), is associated with increased oxygen affinity and reduced homotropic and heterotropic allostereism.

In this paper we describe the consequences of altering the α -chain NH₂ terminus in Hb Thionville. Relative to Hb A, Hb Thionville has a lower affinity for oxygen, and the heterotropic allosteric effects of protons, chloride, and bezafibrate are reduced. The crystal structure of deoxy Hb Thionville provides a basis for understanding the altered functional properties of the mutant hemoglobin.

MATERIALS AND METHODS

Hematologic and Electrophoretic Studies—Hematologic studies were performed by routine methods. Blood samples were collected in

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¹ The abbreviations used are: Hb, hemoglobin; ²R_{symm}, $\sum_{hkl} [|I_{hkl} - I_{hkl}|] / \sum_{hkl} I_{hkl}$; ³R factor = $\sum \|F_o| - |F_c| \| / \sum |F_o|$; σ , standard error/experimental point; 2,3-DPG, 2,3-diphosphoglycerate; bis-Tris, [bis(2-hydroxyethyl)iminotris(hydroxymethyl)methane]; HEPES, 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid; RP-HPLC, reverse-phase high pressure liquid chromatography.

heparin and hemolysates were prepared by lysis of the washed erythrocytes with 4 volumes of water and 0.5 volume of toluene, followed by centrifugation.

The methods for electrophoretic analysis of hemoglobin included: electrophoresis on cellulose acetate at alkaline pH, citrate agar electrophoresis, isoelectric focusing of hemoglobin, electrophoresis of globin chains in urea at pH 6.0 and 9.0 or in the presence of Triton X-100 (Alter *et al.*, 1980; Schneider and Barwick, 1986).

Structural Studies—The abnormal hemoglobin component was isolated by preparative isoelectric focusing on a granulated bed according to a modification of the usual technique (Radola, 1973). Approximately 100 mg of hemolysates were applied to a $26.0 \times 5.0 \times 0.4$ -cm homemade plexiglass cassette. Isoelectric focusing was performed overnight at 4°C at 1300 V and 4 watts.

Globin was prepared just before subunit separation as follows. 200 μL of lysate (50 mg/ml) were precipitated in 3 ml of acid acetone (HCl 0.02 N in acetone) and centrifuged for 5 min at 3000 rpm. The supernatant was removed, the globin pellet was solubilized in 200 μL of water, precipitated with 3 ml of acid acetone, and centrifuged again. The α -chain was separated from the β -chain by reversed-phase high performance liquid chromatography (RP-HPLC) using a Partisil 10 ODS-2 column (25×0.46 cm) (Touzart et Matignon, France) eluted, at a flow rate of 0.8 ml/min, by developing a gradient of isopropyl alcohol in trifluoroacetic acid 0.2% in water. Solvent A and B were made from isopropyl alcohol and trifluoroacetic acid 0.2% in water, respectively, in the ratios 20:80 (v/v) and 60:40 (v/v). The gradient was as follows: from 20 to 35% solvent B in 3 min, from 35 to 50% in 40 min, and from 50 to 60% in 5 min.

After aminoethylation, digestion with trypsin (L-1-tosyl-amido-2-phenylethyl chloromethyl ketone-treated) was performed in 25 mM ammonium bicarbonate, pH 8.8, using an enzyme to substrate ratio of 3:100 (w/w). The peptides were separated by RP-HPLC at a flow rate of 1 ml/min on an Aquapore RP 300 column (25×0.46 cm) (Brownlee Laboratories, Santa Clara, CA). Elution was obtained in 120 min by developing a linear gradient from 0 to 30% acetonitrile in either 50 mM ammonium acetate or 0.05% trifluoroacetic acid. Peptides were manually collected in small Pyrex tubes and dried.

In order to determine their amino acid composition and to prevent methionine degradation, the peptides were hydrolyzed under vacuum for 22 h at 110°C in 40 μL 6 N HCl (Pierce Chemical Co.) in the same tubes. The samples were converted into phenylthiocarbamyl derivatives (phenylthiocarbamyl amino acid) according to the method of Heinrichson and Meredith as previously described (Heinrichson and Meredith, 1984; Baudin and Wajzman, 1987).

Mass Spectrometry—Electrospray ionization experiments were performed using a BIO-Q instrument from VG-Mass Laboratories (Manchester, United Kingdom (U. K.)). The protein (approximately 25 μg) was dissolved into 70 μL of a 1:1 mixture of water and methanol containing 1% acetic acid. 10 μL of this solution were injected into the liquid flow (2 $\mu\text{L}/\text{min}$). Mass scale calibration employed the multiply charged peaks of normal α -globin ($M_r = 15,126.4$) from a separate introduction of normal hemoglobin.

Liquid secondary ion mass spectra were recorded on a ZAB 2E double focusing instrument (VG Analytical, Manchester, U. K.). Ionization was induced by a cesium gun working at 35 kV. The matrix was a 1:1 mixture of glycerol and thioglycerol (1 μL) containing 1% trichloroacetic acid.

B/E linked scans were used for collision-activation-dissociation measurements. Air was the collision gas.

Thermal Stability and Auto-oxidation Rates—The stability of Hb Thionville was tested by determining the thermal stability and auto-oxidation rates. Heat stability was tested by incubating the stripped hemoglobin components (10 mg/ml) at 65°C in 10 mM phosphate buffer, pH 7.0, and measuring the hemoglobin concentration in the supernatant at various time points. Auto-oxidation kinetics were measured spectrophotometrically after incubation of the stripped, purified hemoglobins at 37°C in 50 mM bis-Tris, pH 6.5, with 0.1 M NaCl.

Functional Studies—DEAE-Sephadex chromatography was used to isolate the mutant hemoglobin. A linear pH gradient was developed between pH 7.8 and 6.8 in 50 mM Tris-HCl buffer. The eluant was monitored at 540 nm. The hemoglobin fractions were collected and concentrated by dialysis under vacuum. The purified Hb Thionville was stripped of anions by passing it over a mixed ion-exchanger column.

Oxygen equilibrium curves for the mutant hemoglobin were performed at 25°C with an automatic, continuous method (Hemox Analyzer) as described previously (Kister *et al.*, 1987). The buffer was

either 50 mM bis-Tris or Tris-HCl to which 50 mM sodium EDTA and catalase (20 $\mu\text{g}/\text{ml}$) were added in order to limit methemoglobin formation. Methemoglobin concentrations, as determined using the wavelengths 630, 576.5, and 560 nm according to the method of Benesch *et al.* (1973), never exceeded 5% at the end of the 60-min recording. The hemoglobin concentration was 60–80 μM on a heme basis. As indicated in the figure legends and the tables, the experiments were carried out at different pH values (pH 6.5, 7.2, and 7.8) in the presence of varying concentrations of NaCl (0.1 and 0.4 M). Experiments were also made in the absence of chloride using a 50 mM HEPES buffer at pH 7.2. Additionally, oxygen equilibrium curves were obtained for solution conditions in which either 2,3-DPG or bezafibrate were in excess. Stock solutions of these effectors were prepared in twice-distilled water and adjusted to pH 7.0.

The oxygen equilibrium data were analyzed as previously described (Kister *et al.*, 1987). p_{50} and n_{50} values were calculated by linear regression from the Hill equation for oxygen saturation levels between 40 and 60%. The magnitude of the heterotropic effects was calculated as a $\Delta \log p_{50} \pm$ the effector. For all sets of conditions, the data for Hb Thionville were compared to data for stripped Hb A obtained under identical conditions. The oxygen equilibrium data were also analyzed within the framework of the two-state allosteric model of Monod *et al.* (1965). The allosteric parameters were obtained after fitting the experimental curves to the equation for Y by a non-linear least-squares algorithm according to Bevington (1969). Initial estimates for K_R and K_T , the oxygen dissociation constants for the R and T states, respectively, were obtained from the Hill plots assuming a slope of unity for $Y > 0.99$ and $Y < 0.01$. The quality of the fittings was estimated by statistical analysis to obtain the standard error/experimental point(s). In Table V, L is the allosteric constant, c is the K_T/K_R ratio, Lc^3 is the allosteric equilibrium, and $T_3\%$ represents the percentage of T state hemoglobin at the third ligation step; i'_s is the switchover point ($-\log L/\log c$) indicating the degree of ligation at the $T \rightarrow R$ transition.

X-ray Crystallographic Analysis—Purified Hb Thionville was dialyzed into 10 mM ammonium phosphate, pH 7.0, prior to crystallization. Crystals of deoxy Hb Thionville, grown according to the procedure of Perutz (1968), were isomorphous with deoxy Hb A crystals (space group symmetry $P2_1$ with $a = 63.2$ Å, $b = 83.6$ Å, $c = 53.8$ Å, and $\beta = 99.4^\circ$). Diffraction data were collected with three crystals on a Rigaku AFC6 diffractometer fitted with a San Diego Multiwire Systems area detector. Friedel pairs were collected, resulting in a total of 754,736 measurements. The data were scaled and merged using the method described by Howard *et al.* (1985), yielding 69,608 independent reflections. The data are 93% complete out to a resolution of 1.58 Å and of good quality as judged by an R_{sym} of 5.4% on intensity.

An initial model of deoxy Hb Thionville was constructed from a refined structure of deoxy Hb A structure. Specifically, our structure of deoxy Hb A was obtained by refining the published deoxy Hb A structure (Fermi *et al.*, 1984) against a new 1.5-Å diffraction data set to a R factor of 16.8%.² There are no significant differences between this deoxy Hb A structure and the published deoxy Hb A structure of Fermi *et al.* The acetylated methionine was added to the α -chain NH_2 terminus, and the valine to glutamate substitution was made using the interactive computer graphics program TOM/FRODO (Cambillau, 1989), a version of the program FRODO (Jones, 1985). Deleted $|F_{\text{observed}} - F_{\text{calculated}}|$ Fourier maps in which residues near the mutation site were omitted from the model were used to construct the deoxy Hb Thionville model. A difference electron density map calculated with the known deoxy Hb A phases and $|F_{\text{HbThionville}} - F_{\text{HbA}}|$ magnitudes was also helpful in positioning the NH_2 -terminal glutamate and methionine, as the $|F_{\text{observed}} - F_{\text{calculated}}|$ Fourier maps had weak electron density for these residues.

The structure of deoxy Hb Thionville was refined using the restrained least-squares refinement program PROLSQ of Konnert and Hendrickson (Hendrickson, 1985) running on a VAX 6400. After a total of 37 PROLSQ cycles and five sessions of visual refitting, the R factor for the deoxy Hb Thionville structure was 15.2% for data with intensities greater than 2σ between 6.0 and 1.7 Å resolution. At this point in the refinement, two conformations for the methionine side chain were modeled for the $\alpha 1$ chain. It was not possible to locate multiple conformations for the methionine at the $\alpha 2$ NH_2 terminus due to its greater mobility. The deoxy Hb Thionville model containing the dual conformations for Met (-1) $\alpha 1$ was refined using a modified version of PROLSQ that is capable of handling multiple side chain

² J. S. Kavanaugh and A. Arnone, unpublished results.

conformations (Sheriff, 1987). A total of 20 least-squares cycles, run on a Silicon Graphics 4D-70G, and two manual rebuilding sessions completed the refinement. The final deoxy Hb Thionville model has a standard crystallographic *R* factor of 15.7% for all data with intensities greater than 2σ between 6.0 and 1.5 Å resolution. The refinement statistics for the final cycle are presented in Table VI.

RESULTS

Case Report

The abnormal hemoglobin was identified in a 78-year-old French diabetic patient during routine measurement of glycosylated hemoglobin. The chromatographic assay (Diamat[®]) revealed an aberrant hemoglobin fraction, amounting to 21%, eluting between HbA_{1c} and HbA₀. No hematological abnormalities were observed (red blood cell: 4.17×10^{12} /liter, Hb 13.4 g/deciliter, mean cell volume: 97 femtoliter; reticulocytosis: 1.7%).

Electrophoretic Studies

A comparison of the mobilities of the hemoglobin variant with those of Hb A are given in Table I. Such an electrophoretic pattern, showing the largest difference in migration of the α subunit at acidic pH, suggested the replacement of a protonated residue by an acidic one. In addition, the variant component showed no electrophoretic heterogeneity, which would have resulted from a processing event affecting only a fraction of the mutated hemoglobin, or a stable equilibrium between conformational states that could result in different pI.

Structural Characterization

The mutant hemoglobin was isolated by preparative isoelectric focusing on a granulated bed. The structural analysis included purification of the abnormal α -chain, separation of the tryptic peptides by RP-HPLC, and determination of the amino acid composition of the peptides after phenylisothiocyanate derivatization. The peptide map showed that the α T-1 peptide from Hb Thionville was much more hydrophobic than the normal α T-1 peptide, eluting from the RP-HPLC column with the α T-4 peptide in the trifluoroacetic acid system (Fig. 1, chromatogram a). Purification of the abnormal α T-1 required a second chromatographic separation, using a gradient of acetonitrile in 50 mM ammonium acetate, pH 6.0 (Fig. 1, chromatogram b). The amino acid composition of the α T-1 peptide revealed multiple abnormalities; a valine was missing and 2 residues normally not found in α T-1, a glutamic acid and a methionine, were present (Table II). This amino acid composition suggested that, as observed for Hb Doha (Kamel *et al.*, 1985), replacement of the NA-1 valine by a glutamic acid residue resulted in the α -chain being extended by 1 residue due to failure of the initiator methionine to be removed. Microsequence analysis of this peptide by Edman degradation was not possible, suggesting that the methionine residue was blocked.

Mass Spectrometry

Electrospray Ionization Mass Spectra—The spectrum obtained by electrospray (Covey *et al.*, 1988) of the abnormal α -chain is shown on Fig. 2a. The crude spectrum (not shown) presented a series of major signals due to the several protonated states of the intact protein. The software calculated and transformed this spectrum into a new one using a molecular weight scale. The molecular mass of the α -chain was measured at 15,329.3 Da (the theoretical molecular mass of the normal α -chain of hemoglobin is 15,126.4). The standard deviation was about 0.5 Da. Thus the mass of the modification was estimated to 202.9 ± 1 Da.

Liquid Secondary Ion Mass Spectra Analysis of the α T-1 Peptide—The liquid secondary ion mass spectrum from the abnormal tryptic peptide showed an ion at m/z 932.4 (Fig. 2b). The calculated mass difference between the latter and protonated normal α T-1 (m/z 729.4) is equal to 203.0 ± 0.2 Da, a very close agreement with the mass difference between modified and normal α -chain as measured by electrospray ionization method.

Collision-Activation-Dissociation Mass Spectra—The amino sequence of a peptide can be deduced from the collision-activation-dissociation spectrum of its protonated molecular ion (Bieman, 1988). In that way, the collision-activation-dissociation spectrum of protonated abnormal α T-1 at m/z 932.4 showed several sequence ions. Some of these contained carboxyl terminus part (Y, Z series), others contained the amino terminus part of the molecule (B, C series) (Roepstorff and Fohlman, 1984). (Fig. 2c).

The Y, Z ions series appeared at the same values as from normal α T-1. The B and C ions from the common sequence contained the blocked and modified aminoend part of the peptide. They appeared as shifted up 203 mass units comparatively to the corresponding ones from normal α T-1. However, two additional characteristic and cleavages near the amino end afforded B ions at m/z 174 and 303, that could be attributed to *N*-acetyl-Met and *N*-acetyl-Met-Glu sequences, respectively.

Heat Stability and Auto-oxidation

The thermal stability of Hb Thionville was found to be slightly reduced as compared to that of Hb A and almost the same as that of Hb F.

The rate of auto-oxidation for Hb Thionville is comparable to that of Hb A.

Functional Studies

The purity of the Hb Thionville isolated by DEAE-Sephadex chromatography was checked by both isoelectric focusing and by ion-exchange HPLC of the subunits in 6 M urea. The mutant hemoglobin was found to be more than 95% pure.

The oxygen equilibrium curves of the isolated Hb Thionville showed several abnormalities.

Overall Oxygen Affinity and Cooperativity—In standard solution conditions (pH 7.2, 0.1 M NaCl, 25 °C), Hb Thionville exhibits an almost 2-fold decrease in oxygen affinity compared

TABLE I
Electrophoretic parameters

The isoelectric focusing (IEF) migration is expressed in mm from the position of Hb A; the other parameters were determined according to a comparative scale as described by Schneider and Barwick (1986).

Hemoglobin	IEF	Cellulose acetate alkaline pH	Citrate agar	pH 9.0	6 M urea, pH 6.0, X-100	Triton
Hb A	0.0	0.0	0.0	10.0	10	10
Hb Thionville	+6.8 mm	3.7	0.0	12.1	16.4	13.1

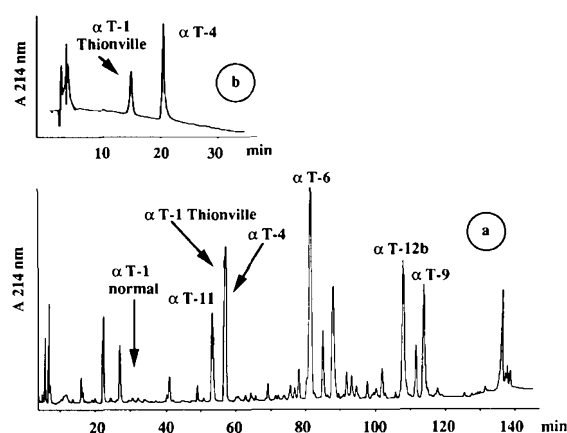


FIG. 1. RP-HPLC peptide analysis. *panel a*, elution pattern of the tryptic digest of the α -chain from Hb Thionville. Peptide α T-1 was absent at its normal position and eluted with increased retention time together with peptide α T-4. Experimental conditions: 0.5 mg of tryptic digest were applied on a Aquapore RP 300 column, and elution was achieved at a flow rate of 1 ml/min by a linear gradient of acetonitrile from 0 to 25% in trifluoroacetic acid 0.05%. *panel b*, purification of peptide α T-1: the peak containing peptides α T-1 and α T-4 was further purified by chromatography on the same column eluted with a gradient of acetonitrile from 0 to 25% in 10 mM ammonium acetate, pH 6.0, developed in 60 min.

TABLE II

Amino acid composition of peptide α T-1 of Hb Thionville

Amino acid	α T-1 Hb Thionville	α T-1 Hb A
Asp	1.1	1.0
Glu	1.1	
Ser	1.1	1.0
Ala	1.1	1.0
Pro	1.1	1.0
Val		1.0
Met	0.9	
Leu	0.9	1.0
Lys	0.9	1.0

to Hb A with a reduced n_{50} value (Table III). The low oxygen affinity of Hb Thionville is pH dependent, being more pronounced at alkaline pH than in the acid pH range. This indicates a reduction of the alkaline Bohr effect of approximately 40% (Table IV). Because chloride binding and the alkaline Bohr effect are linked phenomena at the α -chain NH_2 terminus, the loss of the latter should lead to loss or reduction of the former. The results given in Table IV demonstrate that this is indeed the case. The allosteric effect of chloride is reduced by 44% in Hb Thionville. By contrast, the effect of 2,3-DPG on the oxygen affinity of Hb Thionville is similar to that observed for Hb A (Table IV). This indicates normal 2,3-DPG binding to the Hb Thionville β -chains. The alteration of the chloride effect in Hb Thionville can therefore be assigned to the α -chain-binding sites. The abnormal functional properties of the α -chains in Hb Thionville were confirmed by the observation of a large reduction in the effect of bezafibrate (Tables III and IV), whose binding sites are mainly located at the $\alpha 1$ - $\alpha 2$ interface (Perutz, *et al.*, 1986). Fig. 3 illustrates these results.

Allosteric Behavior of Hb Thionville—In the absence of chloride (HEPES buffer), Hb Thionville exhibits an almost 3-fold decrease in oxygen affinity compared to Hb A (Table III). Analyses of the oxygen binding curves within the framework of the two-state allosteric model are shown in Table V. These results demonstrate that the low oxygen affinity of Hb Thionville in the absence of anionic effectors can be mainly attributed to a high value of K_T , indicating that the allosteric

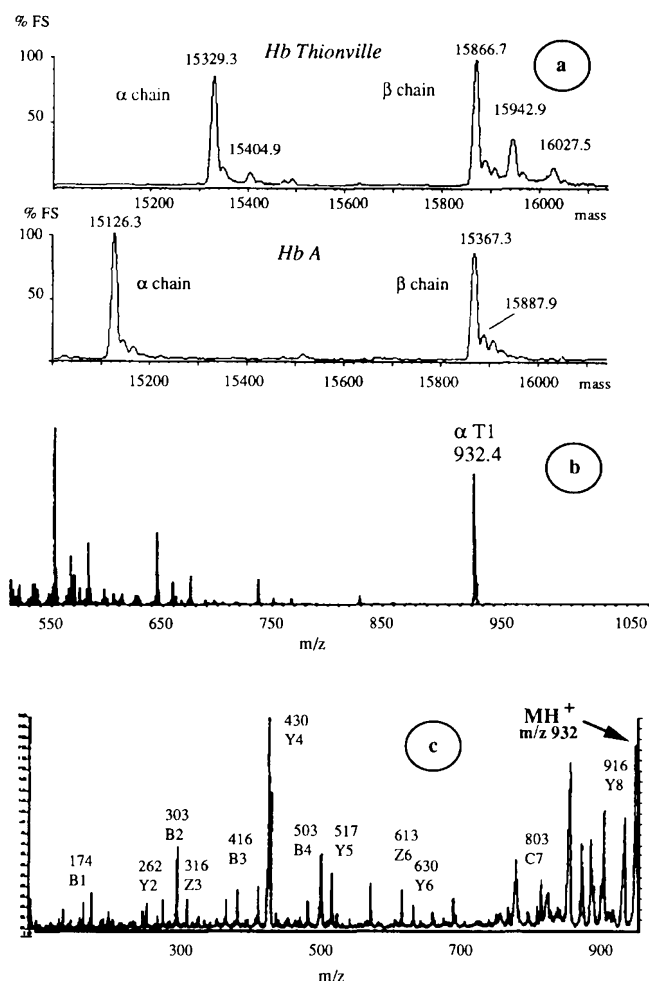


FIG. 2. *Panel a*, electrospray mass spectrum from the mixture of α - and β -chain. The crude spectrum was constituted by series of peaks which correspond to both proteins with different protonation states. Computer calculations transformed the mass over charge scale into a true mass scale showing the calculated molecular weight of the proteins. *Panel b*, liquid secondary ion mass spectrum from abnormal tryptic peptide. The matrix was glycerol/thioglycerol 1:1 containing 1% trichloroacetic acid. *Panel c*, collision-induced-dissociation mass spectrum of protonated abnormal α T-1 (*B/E*-linked scans). The fragment ions were named according to the Roepstorff's nomenclature.

equilibrium is shifted toward the *T* state. This is shown in Table V, which gives the values of the allosteric equilibrium at the third step in ligation (Lc^3). This indicates that the structural abnormality in Hb Thionville leads to more *T* state, in agreement with the diminished slope of the Hill plot (Fig. 3). In addition, the values in the absence of chloride also indicate that the *T* state in Hb Thionville is different from the *T* state in Hb A. This is born out by the observation that addition of 100 mM Cl^- leads to a 6-fold increase in K_T for Hb A and less than a 2-fold increase for Hb Thionville, indicating a lower intrinsic affinity for the mutant hemoglobin. In other words, Hb Thionville behaves as if it contains an internal structural feature acting in a similar fashion as an allosteric effector. Interestingly, the value of K_T for Hb Thionville is barely sensitive to the presence of bezafibrate, contrasting with the normal allosteric effect of 2,3-DPG. This demonstrates a clear differentiation in Hb Thionville between the functional properties of the α - and β -chains in the presence of their specific effectors.

TABLE III
Oxygen binding properties of Hb Thionville and Hb A

Experimental conditions	p_{50}	n_{50}	$\frac{p_{50} \text{ Hb Thionville}}{p_{50} \text{ Hb A}}$
	torr		
pH 7.2 $[\text{Cl}^-] = 0 \text{ M}$			
Hb Thionville	5.5	2.3	2.7
Hb A	2.0	2.5	
pH 7.2 $[\text{Cl}^-] = 0.1 \text{ M}$			
Hb Thionville	9.7	2.4	1.8
Hb A	5.4	3.0	
pH 7.2 $[\text{Cl}^-] = 0.4 \text{ M}$			
Hb Thionville	14.3	2.1	1.2
Hb A	12.0	2.3	
pH 6.4 $[\text{Cl}^-] = 0.1 \text{ M}$			
Hb Thionville	15.7	2.4	1.3
Hb A	12.2	2.7	
pH 7.8 $[\text{Cl}^-] = 0.1 \text{ M}$			
Hb Thionville	6.1	2.1	2.6
Hb A	2.3	2.7	
pH 7.2 $[\text{Cl}^-] = 0.1 \text{ M}$; DPG = 2 mM			
Hb Thionville	27.1	2.1	1.7
Hb A	15.9	2.9	
pH 7.2 $[\text{Cl}^-] = 0 \text{ M}$; beza- fibrate = 3 mM			
Hb Thionville	12.8	2.4	0.8
Hb A	16.7	2.5	

TABLE IV
Heterotropic effects at half-saturation (p_{50}) for Hb Thionville and Hb A

	Hb Thionville	Hb A
Chloride effect		
$\Delta \log p_{50} \pm 0.1 \text{ M Cl}^-$	0.25	0.43
Bezafibrate effect ($[\text{Cl}^-] = 0 \text{ M}$)		
$\Delta \log p_{50} \pm 3 \text{ mM Bezafibrate}$	0.37	0.93
DPG effect ($[\text{Cl}^-] = 0.1 \text{ M}$)		
$\Delta \log p_{50} \pm 2 \text{ mM 2,3-DPG}$	0.45	0.46
Bohr effect (pH 6.4 $\rightarrow$ pH 7.8)		
$\Delta \log p_{50} / \Delta \text{pH}$	-0.31	-0.56

X-ray Diffraction Studies

Because the mutation-induced structural changes are localized to the mutation site, the deoxy Hb Thionville crystal structure is highly isomorphous with that of deoxy Hb A. The largest main chain atomic displacements occur at the NH_2 termini of the α -chains and have magnitudes of $\sim 0.3 \text{ \AA}$ for the $\alpha 1$ subunit and $\sim 0.6 \text{ \AA}$ for the $\alpha 2$ subunit. Elsewhere, the differences in main atomic coordinates never exceed $0.15 \text{ \AA}$.

The high degree of structural isomorphism between deoxy Hb A and deoxy Hb Thionville is evident in Fig. 4A, which illustrates the spatial relationship between the NH_2 terminus of the $\alpha 1$ subunit and the COOH terminus of the $\alpha 2$ subunit for deoxy Hb Thionville (*thin bonds*) and deoxy Hb A (*thick bonds*). With the exception of the additional acetylated methionine and the glutamate side chain, the structure remains essentially unchanged. The only significant main chain atomic displacements, entail the movement of the NH_2 -terminal dipeptide (Glu-1 and Leu-2) away from the COOH -terminal region of the other α -chain (Fig. 4A). Despite this shift, the network of interactions involving direct contacts between the α -chains (Fermi *et al.*, 1984) remains undisturbed. The intrasubunit interaction between the ϵ -amino

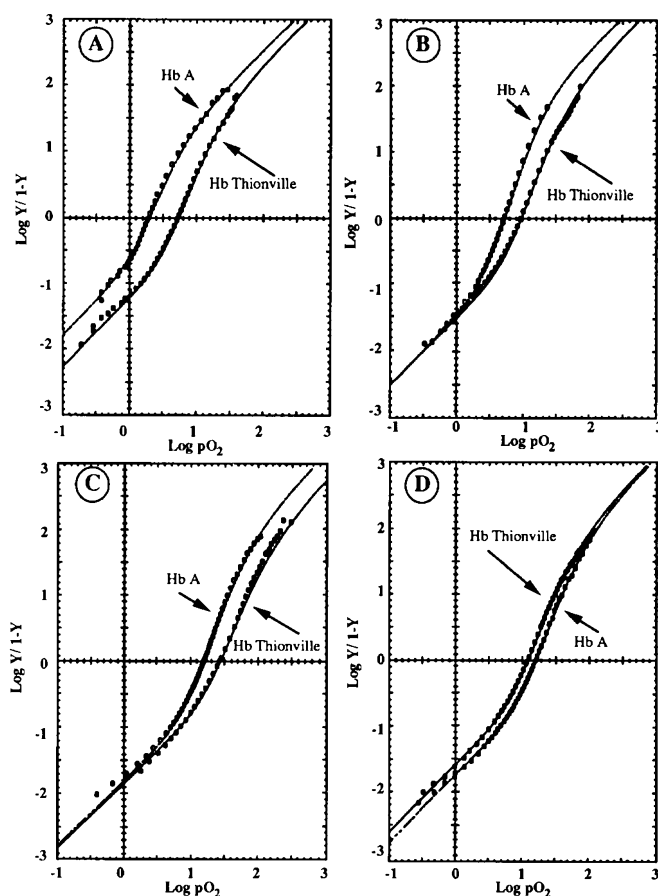


FIG. 3. Oxygen binding curves for Hb Thionville and Hb A. Panel A, HEPES 50 mM, pH 7.2, 25 °C; panel B, bis-Tris 50 mM, NaCl 0.1 M, pH 7.2, 25 °C; panel C, bis-Tris 50 mM, NaCl 0.1 M, 2,3-DPG 2 mM, pH 7.2, 25 °C; panel D, HEPES 50 mM, bezafibrate 3 mM, pH 7.2, 25 °C. Symbols are one of four experimental data points, lines are non-linear least-square fits using the MWC model parameters given in Table V.

group of Lys-127 α and carbonyl oxygen of Val-1 α is maintained; the ϵ -amino group of Lys-127 α shifts slightly toward Glu-1 α so that it remains within hydrogen bonding distance to the carbonyl oxygen (Fig. 4A). Although the pattern of direct interactions that constitute the $\alpha 1$ - $\alpha 2$ interface is not disrupted in deoxy Hb Thionville, the valine to glutamate substitution and the NH_2 -terminal acetylated methionine do modify the $\alpha 1$ - $\alpha 2$ interface by forming new interactions and by disrupting the binding of heterotropic ligands within the interface.

The effect of the valine to glutamate substitution on the structure of $\alpha 1$ - $\alpha 2$ interface is minimal or indirect because the glutamate side chain makes no direct interactions with residues in the interface. The side chain of Glu-1 α extends away from the interface into the bulk solvent (Fig. 4A). The carboxyl group does not participate in any strong charge interactions; the nearest positive charge is the amino group of Lys-7 $\alpha 1$, which is nearly $6 \text{ \AA}$ away. As a result, the carboxyl group is very mobile, which is indicated by a temperature factor of $73.6 \text{ \AA}^2$ and weak density in the deleted $|F_{\text{observed}} - F_{\text{calculated}}|$ Fourier map (Fig. 4C). The β and γ carbons have somewhat more density and lower temperature factors of 34.5 and $58.23 \text{ \AA}^2$, respectively. On the $\alpha 2$ chain Glu-1 is even more mobile, having an average side chain temperature factor of $99.0 \text{ \AA}^2$.

The impact of the NH_2 -terminal acetylated methionine on the structure of the $\alpha 1$ - $\alpha 2$ interface is much more apparent.

TABLE V
Allosteric parameters for Hb Thionville and Hb A

σ was in the range of $3\text{--}5 \times 10^{-3}$ for all measurements. $i'_s = -\log L/\log c$ (switch over point); $\%T_3 = Lc^3/(1 + Lc^3)$.

Hb	L	c	K_R	K_T	Lc^3	$\%T_3$	i'_s
[Cl ⁻] = 0 M, pH 7.2							
Hb Thionville	5.1×10^4	0.019	0.33	17.3	0.35	26	2.7
Hb A	3.8×10^3	0.039	0.24	6.2	0.22	18	2.5
[Cl ⁻] = 0.1 M, pH 7.2							
Hb Thionville	2.7×10^5	0.011	0.39	33.6	0.36	26	2.8
Hb A	1.4×10^5	0.008	0.27	33.3	0.07	7	2.4
[Cl ⁻] = 0 M; [bezafibrate] = 3 mM, pH 7.2							
Hb Thionville	3.7×10^5	0.012	0.47	39.6	0.64	39	2.9
Hb A	8.3×10^5	0.009	0.50	56.3	0.66	40	2.9
[Cl ⁻] = 0.1 M, [2,3-DPG] = 2 mM, pH 7.2							
Hb Thionville	1.9×10^6	0.009	0.65	71.6	1.45	60	3.0
Hb A	6.5×10^5	0.008	0.52	65.0	0.33	25	2.8

The methionine side chain, unlike the glutamate, protrudes into the $\alpha 1\text{--}\alpha 2$ interface (Fig. 4A). Two side chain conformations were identified for Met (-1) of the $\alpha 1$ chain (Fig. 4A); a major conformation referred to as rotomer 1 (*Rot 1*) and a minor conformation referred to as rotomer 2 (*Rot 2*). Only one conformation corresponding to the major rotomer was identified for Met (-1) in the $\alpha 2$ subunit (data not shown). Rotomer 1 has side chain χ angles of $\chi 1 = -10^\circ$, $\chi 2 = -105^\circ$, and $\chi 3 = -144^\circ$, and rotomer 2 has χ angles of $\chi 1 = 156^\circ$, $\chi 2 = -148^\circ$, and $\chi 3 = 64^\circ$, values that deviate significantly from the ideal staggered values of $\pm 60^\circ$ or 180° (Ponder and Richards, 1987). These deviations from ideal values may be due to multiple conformations of the methionine main chain atoms and the acetyl group which were not included in the model. Additionally, there may be more than two conformations for the methionine side chain with the C_β and C_γ atoms, in particular, assuming multiple positions. However, it is clear from the electron density images (Fig. 4, B and C) that the methionine side chain is centered in density, and, therefore, the atomic coordinates represent average atomic positions. Better electron density for rotomer 1 than rotomer 2 in Fig. 4, B and C, indicates that rotomer 1 is either more occupied or less mobile than rotomer 2. When the occupancy of the two rotomers was set to 0.5, the average side chain temperature factors were $34.6 \text{ \AA}^2$ for rotomer 1 and $52.1 \text{ \AA}^2$ for rotomer 2.

The intrusion of the methionine side chain into the $\alpha 1\text{--}\alpha 2$ interface results in the formation of several new intrasubunit and intersubunit contacts. These interactions can be seen in Fig. 4A and are depicted schematically in Fig. 5. Rotomer 1 and the acetyl group make multiple contacts that span the $\alpha 1\text{--}\alpha 2$ interface. While most of these interactions are weak van der Waals contacts ($3.25\text{--}4.0 \text{ \AA}$ in length), two are potential hydrogen bonds. One potential hydrogen bond consists of an interaction between the carbonyl oxygen of Ser-138 $\alpha 2$ and the main chain methionine N. The second involves a water molecule (referred to as Sol 2), found in both the mutant and the native models, that interacts with the oxygen of the acetyl group and the hydroxyl of Thr-137 across the $\alpha 1\text{--}\alpha 2$ interface. The distances of these two interactions are about $3.4 \text{ \AA}$, indicating that they may be weak hydrogen bonds. Rotomer 2 does not contribute any new intersubunit contacts, but does make several new intrasubunit contacts with residues on the H helix (Fig. 5).

In addition to contributing new interactions that may be involved in stabilizing deoxy Hb Thionville, the position of the NH_2 -terminal methionine accounts for the decreased re-

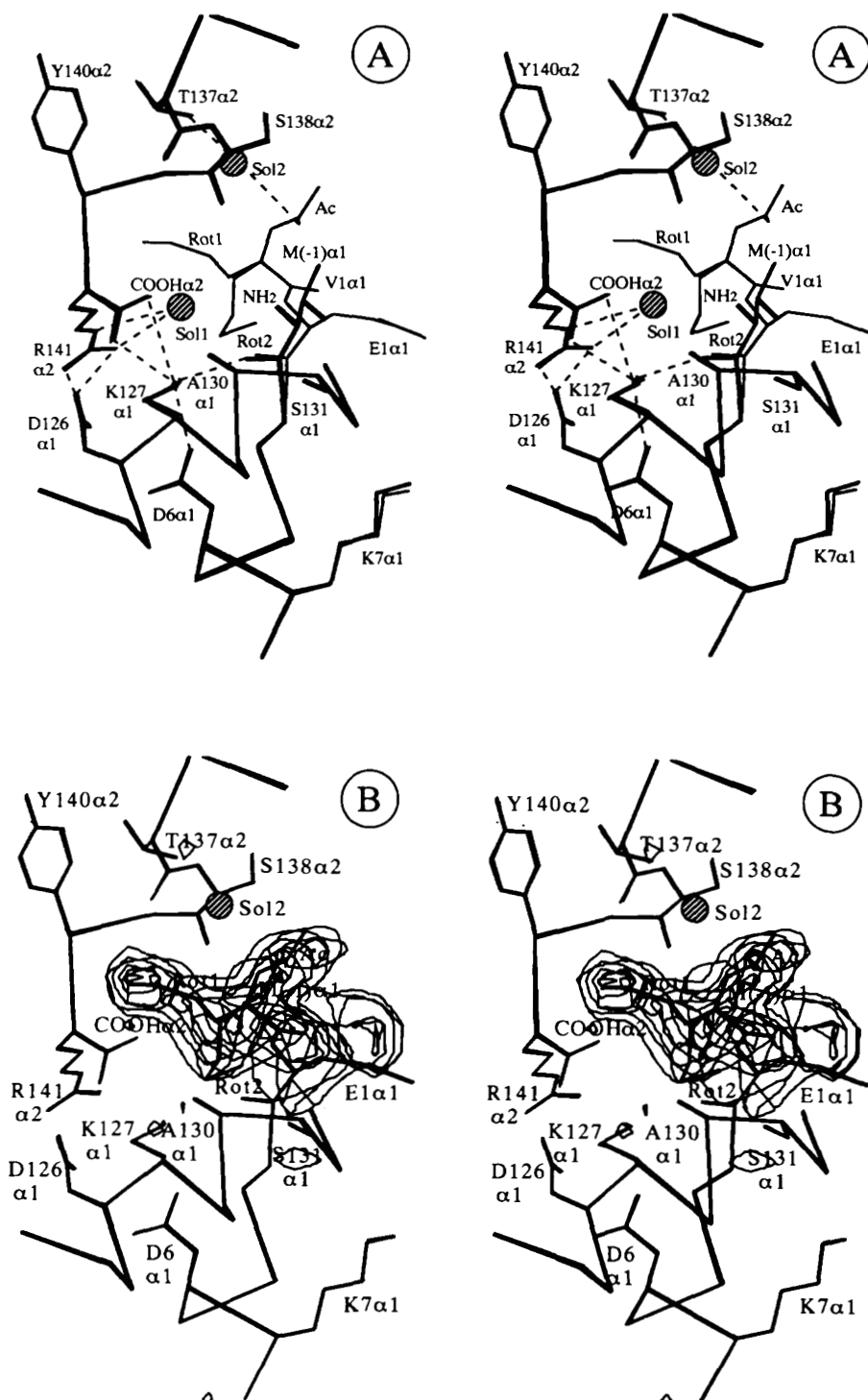
sponses to bezafibrate and chloride. The methionine side chain physically displaces the effector ligands by either occupying a portion of the bezafibrate-binding site or by making unacceptably close contacts with chloride. Bezafibrate binds in the central cavity with contacts to one β -chain and both α -chains. The isobutyl group of bezafibrate is between Arg-141 and Thr-137 of the same α -chain (Perutz, *et al.*, 1986), which is the same location occupied by the sulfur and ϵ -methyl group of the major methionine rotomer (Figs. 4A and 5).

The reduced chloride effect correlates with the elimination of the anion-binding sites associated with the NH_2 terminus of the α -chain. X-ray crystallographic analysis of deoxy Hb A has identified two sites near the NH_2 terminus of each α -chain that are capable of binding sulfate or phosphate (Nigen *et al.*, 1976; Arnone and Williams, 1977; O'Donnell *et al.*, 1979; Fantl *et al.*, 1987). Results from solution studies indicate that these same sites bind chloride (Chiancone *et al.*, 1975; Nigen *et al.*, 1976; O'Donnell *et al.*, 1979; Fantl *et al.*, 1987). One site is located between the α -amino group of Val-1 α and the β -hydroxyl of Ser-131 on the same α -chain, and the other site is located between the α -amino group of Val-1 α and the guanidinium group of Arg-141 of the opposite α -chain. This solvent peak is labeled Sol 1 in Fig. 4, a and d. The negative peak at the position of Sol 1 in the deoxy Hb Thionville - deoxy Hb A difference map (Fig. 4D), and the absence of a solvent peak at this site in the deoxy Hb Thionville Fourier map indicates that this anion-binding site is lost as a result in deoxy Hb Thionville. Acetylation of NH_2 terminus and the resulting loss of the positive charge is probably a contributing factor to the loss of the Sol 1 site. However, close contacts between a chloride ion at the Sol 1 site and the methionine side chain in its rotomer 2 conformation may be the main reason for the elimination of the binding site. A negative difference electron density peak between the NH_2 terminus and Ser-131 in the deoxy Hb Thionville - deoxy Hb A difference map indicates that the occupancy of the second anion-binding site is also reduced (Fig. 4D).

DISCUSSION

The phenotypic expression of a hemoglobin variant depends not only on the amino acid substitution, but also on several other features, as for example alteration in the mechanism of processing and post-translational modification of the protein structure. In Hb Thionville, the modified primary sequence inhibits the release of the NH_2 -terminal methionine, which in turn allows acetylation of the NH_2 -terminal amino group

FIG. 4. Stereo plots showing the relationship of the NH₂ terminus of the α 1 subunit and the COOH terminus of the α 2 subunit. Panel A, comparison of the deoxy Hb Thionville structure (thin bonds) with the deoxy Hb A structure (thick bonds). The dashed lines indicate the network of interactions between Val-1, Asp-6, Asp-126, Lys-127 of one α -chain and Arg-141 of the other α -chain. The two conformations for Met (-1) are designated Rot 1 and Rot 2. Panel B, positive contours of the (deoxy Hb Thionville - deoxy Hb A) difference electron density map superimposed on the deoxy Hb Thionville model. Contours are drawn at increments of three times the root-mean-square value of the map. Panel C, $|F_{\text{observed}} - F_{\text{calculated}}|$ Fourier map, in which Met (-1) and Glu-1 have been omitted from the model, superimposed on the deoxy Hb Thionville model. Contours are drawn at increments of 1.5 times the root-mean-square value of the map. Panel D, negative contours of (deoxy Hb Thionville - deoxy Hb A) difference electron density map superimposed on the deoxy Hb A model. Contours are drawn at increments of three times the root-mean-square value of the map.



to occur. A few hemoglobin variants in which the initiator methionine is retained have been reported in the β -chain, but Hb Thionville is the first one concerning the α -chain.

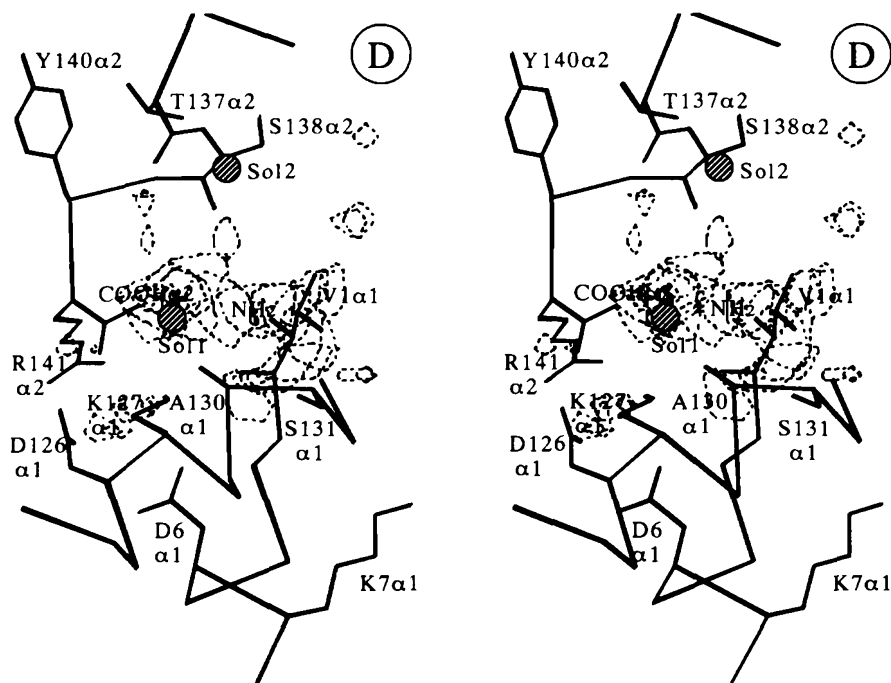
Hb Thionville is an additional example showing that, in red blood cells, the release of the initiator methionine is fully prevented by the presence of a glutamic acid at the second position of the nascent polypeptide chain. This situation is also observed for other erythrocytic proteins having the same NH₂-terminal sequence, as for example the band 3 protein (Drickamer, 1978). This NH₂-terminal sequence, Met-Glu, appears to be a target for complete acetylation.

In eucaryotic cells, two enzymes are likely to be involved in N^a-acetylation of cytosolic proteins. The first is a methionine

N^a-acetyltransferase with high activity that requires methionine at the NH₂ terminus and an aspartic acid, glutamic acid, or asparagine at position two (Lee *et al.*, 1990). The second enzyme is a N^a-acetyltransferase with broader specificity; alanine, serine, methionine, glycine, and threonine can all be acetylated by this enzyme (Lee *et al.*, 1990). In humans, the effect of the first enzyme is illustrated by hemoglobin variants which have retained their initiator methionine and have a glutamic acid replacing the valine (Hb Doha and Hb Thionville). The action of the second enzyme is illustrated by several examples: 15–20% of the protein is acetylated in Hb South Florida (Boissel *et al.*, 1985), where the NH₂-terminal sequence of the β -chain is Met-Met, and in Hb F where the



FIG. 4—continued



NH₂-terminal residue of the γ -chain is glycine. Conversely, in the β -chain of Hb Raleigh, the NH₂-terminal alanine is fully acetylated (Moo-Penn *et al.*, 1977). The ζ embryonic chain is the only example in humans in which an α -type chain is acetylated.

In Hb Marseille-Long Island (Blouquit *et al.*, 1984; Barwick *et al.*, 1985) the methionine is also preserved, but the β -chain is not acetylated.

Hb Thionville is also of interest because it permits a clear separation of the responses of α - and β -chains to heterotropic effectors. This is demonstrated by the results in Tables III and IV which show Hb Thionville to have reduced responses

to protons, chloride, and bezafibrate, but a normal response to 2,3-DPG. The absence of any global structural changes, either quaternary structure rearrangements or long range tertiary deformations, in Hb Thionville is consistent with the normal response to 2,3-DPG whose binding site is approximately 40 Å away from the mutation site.

The modification of the α -chain NH₂ terminus in Hb Thionville results in decreased oxygen affinity, reduced cooperativity (Fig. 3, Table III), a shift in the allosteric equilibrium toward the T state (Table V), and reduced heterotropic responses (Table IV). The difference in oxygen affinities is greatest in the absence of chloride (Table III), where K_7 for

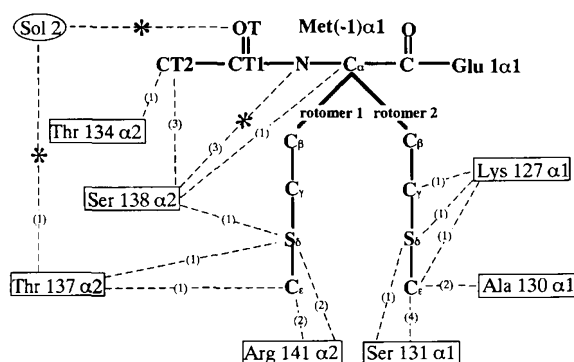


FIG. 5. Scheme showing the intersubunit and intrasubunit contacts made by the acetylated methionine. The dashed lines represent atomic interactions of 4 Å or less. The number in parentheses represents the number of atoms in a particular residue that interact with a particular methionine atom. Possible hydrogen bonds are indicated by an *.

TABLE VI

Summary of least-squares refinement parameters (Hendrickson, 1985)

Parameter	Target σ^b	r.m.s. Δ^a Deoxy Hb Thionville
Bonding distance (Å)		
Bond length (1-2 neighbors) ^c	0.010	0.016
1-3 neighbor distances ^d	0.015	0.031
1-4 planar neighbor distances ^e	0.030	0.054
Planar groups		
Deviations from plane (Å)	0.010	0.015
Chiral centers		
Chiral volumes (Å ³)	0.080	0.162
Non-bonded contacts (Å)		
Separated by one torsion angle	0.200	0.165
All other van der Waals contacts	0.200	0.169
Possible hydrogen bonds	0.200	0.134
Conformational torsion angles (°)		
Planar (e.g., peptide ω)	5.0	3.9
Staggered (e.g., aliphatic χ)	15.0	20.8
Transverse (e.g., aromatic χ_2)	25.0	32.4
Isotropic temperatures (Å ²)		
Main chain (1-2 neighbors) ^c	1.5	2.7
Main chain (1-3 neighbors) ^d	2.0	3.8
Side chain (1-2 neighbors) ^c	4.0	7.9
Side chain (1-3 neighbors) ^d	6.0	11.1

^a r.m.s. Δ , root-mean-square deviation from ideal values as determined from accurate small molecule crystal structures in the case of bonding distances, chiral volumes, and non-bonded contacts, or from average values in the case of isotropic temperature factors.

^b Target σ , estimated standard deviations, where $1/\sigma^2$ is used as a relative weighting factor in the minimized sum of observational functions.

^c 1-2 neighbors, covalently bonded atom pairs.

^d 1-3 neighbors, atom pairs separated by two covalent bonds.

^e 1-4 planar neighbors, atom pairs in a planar group separated by three covalent bonds.

Hb Thionville is almost three times as large as the K_T for Hb A (Table V). When the chloride concentration is increased to 100 mM, K_T is equivalent for the two hemoglobins and the difference in oxygen affinity is attributable largely to a shift in the allosteric equilibrium. These observations suggested that the structural modifications in Hb Thionville may have altered the oxygen affinity of the tetramer in a fashion similar to chloride. In this regard, it is interesting to compare Hb

Thionville with hemoglobins in which the α -amino group of Val-1 α has been modified by carbamylation (Manning *et al.*, 1978; O'Donnell *et al.*, 1979) or carboxymethylation (DiDonato *et al.*, 1983; Fantl *et al.*, 1987).

There are similarities between the functional properties of Hb Thionville and the properties of both carbamylated and carboxymethylated hemoglobins, but there are also important differences between the solution properties of these variant hemoglobins. Blocking of the NH_2 -terminal amine by acetylation in Hb Thionville, or by reaction with cyanate in carbamylated hemoglobin, eliminates the positive charge that is normally associated with α -amino group of Val-1 α . In carbamylated hemoglobin, this leads to the loss of the chloride-binding sites that are associated with the α -chain NH_2 terminus and a large reduction in the chloride-induced alkaline Bohr effect (O'Donnell *et al.*, 1979). The chloride-induced allosteric properties of Hb Thionville, where the α -chain NH_2 -terminal amino group is acetylated, are very similar. However, unlike carbamylated hemoglobin which has increased oxygen affinity (Manning *et al.*, 1978), in Hb Thionville new intersubunit contacts stabilize the *T* state and reduced the oxygen affinity.

In contrast, the oxygen affinity of carboxymethylated hemoglobin more closely resembles that of Hb Thionville. That is, its oxygen affinity is lower than that of hemoglobin A (DiDonato *et al.*, 1983; Fantl *et al.*, 1987). However, in terms of heterotropic effects, carboxymethylated hemoglobin is significantly different from Hb Thionville in that the secondary amine that results from carboxymethylation is still capable of binding both protons and chloride in an oxygen-linked manner (DiDonato *et al.*, 1983; Fantl *et al.*, 1987).

Comparison of the solution properties of Hb Thionville with those of carboxymethylated hemoglobin lead us to speculate that Glu-1 α would be responsible for the altered functional properties. Specifically, in carboxymethylated hemoglobin (Fantl *et al.*, 1987), the carboxyl group occupies the binding site normally found between the α -amino group of Val-1 α and the β -hydroxyl of Ser-131 α . In this position, the carboxymethyl moiety displaces the inorganic anion normally bound at this site and acts as a permanently bound anion that stabilize the deoxy quaternary structure. We expected, therefore, that the carboxyl of Glu-1 α would behave in a similar manner by interacting with Ser-131 α or even with the guanidinium group of Arg-141 on the other α -chain. In this way, the Val to Glu substitution could simultaneously reduce the chloride effect while providing new intersubunit interactions that would stabilize the *T* state and lead to lower oxygen affinity.

This, however, did not prove to be the case as the side chain of Glu-1 α extends away from the $\alpha 1$ - $\alpha 2$ interface in deoxy Hb Thionville. To our surprise, the carboxyl does not participate in any strong polar or charge interactions; the nearest positively charged group being the ϵ -amino group of Lys-7 α which is almost 6 Å away. The altered functional properties of Hb Thionville are due instead to the acetylated methionine. The methionine side chain physically displaces both chloride and bezafibrate (Fig. 4). The increased stability of the *T* state in Hb Thionville (Tables V and VI) is the result of new intrasubunit and intersubunit contacts made by the acetylated methionine (Fig. 5). However, the relative importance of each of the individual interactions that are formed is difficult to estimate because a few indirect interactions through solvent molecules in Hb A have been replaced by several direct non-polar interactions in Hb Thionville.

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